

What is React?

React is a JavaScript library developed by Facebook for building fast and interactive user interfaces using a component-based architecture.

What is JSX?

JSX is a syntax extension for JavaScript that allows writing HTML-like code within JavaScript files. It gets compiled to `React.createElement` calls.

What are components in React?

Components are reusable building blocks in React. They can be functional or class-based and return UI elements.

What is the virtual DOM?

The virtual DOM is an in-memory representation of the real DOM. React updates it first, compares changes, and efficiently updates only what changed in the actual DOM.

What is state in React?

State is a mutable object used to store dynamic data within a component. Updating state re-renders the component.

What are props?

Props are read-only inputs passed from parent to child components, enabling component customization.

Difference between state and props?

State is mutable and managed within a component; props are immutable and passed from parent to child.

What are hooks in React?

Hooks are functions that let functional components use features like state, lifecycle, and context.

What is useState?

`useState` is a hook that manages local state in functional components.

What is useEffect?

`useEffect` manages side effects like API calls, timers, and subscriptions in functional components.

What is useRef?

`useRef` stores a mutable reference that persists across renders and accesses DOM elements.

What is useMemo?

useMemo optimizes performance by memoizing expensive calculations.

What is useCallback?

useCallback memoizes a function definition to prevent unnecessary re■creations.

What is React Router?

React Router is a library used for routing and navigation in React applications.

What is Redux?

Redux is a predictable state management library that stores app state in a central store.

What is a reducer in Redux?

A reducer is a pure function that takes the current state and an action and returns a new state.

What is context API?

Context API allows sharing data across components without passing props manually at every level.

What is lifting state up?

Lifting state up means moving shared state to a common parent to synchronize multiple child components.

What is prop drilling?

Prop drilling is passing props through multiple nested components unnecessarily.

What is React.memo?

React.memo is a higher-order component that memoizes functional components to prevent unnecessary re-renders.

What are keys in React lists?

Keys uniquely identify list items to help React optimize rendering.

Why shouldn't we use index as key?

Using index can cause issues when items re-order, leading to incorrect UI updates.

What is reconciliation?

Reconciliation is React's process of updating the virtual DOM and applying minimal changes to the real DOM.

What are controlled components?

Controlled components use React state to manage form input values.

What are uncontrolled components?

Uncontrolled components store form data in the DOM instead of React state.

What is `useLayoutEffect`?

`useLayoutEffect` runs synchronously before the browser repaints, useful for measuring layout.

What is the difference between `useEffect` and `useLayoutEffect`?

`useEffect` runs asynchronously after paint; `useLayoutEffect` runs synchronously before paint.

What is code-splitting?

Code-splitting breaks bundles into smaller chunks for faster load times.

What is lazy loading in React?

Lazy loading dynamically loads components only when needed using `React.lazy`.

What is `Suspense`?

`Suspense` shows fallback UI while a component is being lazily loaded.

What are higher-order components (HOC)?

HOCs are functions that take a component and return an enhanced component.

What is the difference between functional and class components?

Functional components are simpler and use hooks; class components use lifecycle methods and this keyword.

What are lifecycle methods?

Lifecycle methods like `componentDidMount` run at specific phase of a class component's existence.

What is `Fragment` in React?

Fragments let you group child elements without adding extra DOM nodes.

What is reconciliation algorithm (diffing)?

It compares the old virtual DOM with the new one and updates only changed nodes.

What is the purpose of `useReducer`?

useReducer manages complex state transitions using reducer functions.

What is forwardRef?

forwardRef passes a ref from parent to child components.

What is strict mode in React?

StrictMode highlights potential problems like unsafe lifecycle methods.

What is hydration in React?

Hydration attaches React event listeners to server-rendered HTML.

What is server-side rendering (SSR)?

SSR renders the initial HTML on the server for faster initial loading and SEO.

What is client-side rendering (CSR)?

CSR renders content in the browser using JavaScript, typical for SPAs.

What is React Fiber?

Fiber is React's reconciliation engine enabling asynchronous rendering.

What is an error boundary?

Error boundaries catch JavaScript errors in child components and display fallback UI.

How do you optimize React performance?

Use memoization, code-splitting, virtualization, and avoid unnecessary renders.

What is tree shaking?

Tree shaking removes unused code during bundling, reducing bundle size.

What is custom hook?

Custom hooks allow extracting reusable logic from components.

What is React Portal?

Portals let you render children into a DOM node outside the parent hierarchy.

What is event bubbling?

Event bubbling means events propagate from child to parent elements.